



PROJECT PROPOSAL & TECHNICAL APPROACH

Secure Field Operations Mobile Platform ("DeMars FieldOps Link")

Prepared For:	Kyle Morris, President & Owner DeMars & Associates LTD
Prepared By:	GS3 Solutions Technical Architecture Team
Project Name:	DeMars FieldOps Link (Secure Transfer Agent Portal)
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1. Executive Summary

DeMars & Associates LTD is a nationally recognized alternative dispute resolution (ADR) firm. To execute warranty buyback agreements and Lemon Law dispute resolutions, DeMars deploys on-site Transfer Agents (vehicle retitling specialists) to retail dealerships or consumer locations. These agents are tasked with executing crucial steps: verifying vehicle identification and mileage, auditing cosmetic/structural conditions, obtaining owner signatures on retitling forms/vehicle titles, and exchanging legal manufacturer buyback checks.

GS3 Solutions proposes DeMars FieldOps Link, a custom-designed, highly secure mobile application specifically built for Transfer Agents in the field. The platform streamlines inspection logging, ensures title documents are validly executed, archives photographic proofs, and bridges real-time communication with the DeMars Home Office. Built with an offline-first architecture and military-grade encryption, DeMars FieldOps Link guarantees complete data integrity and compliance, even in areas with zero cell coverage.

2. Platform Architecture & Tech Stack

To ensure portability, rapid updates, and enterprise security, the mobile app and cloud-backend are structured using modern, standardized frameworks:

Architectural Layer	Selected Technology Stack	Business Rationale
Mobile Client	React Native or Flutter	Cross-platform consistency on iOS and Android. High performance with native device access (camera OCR, GPS, SQLCipher).
Local Storage	SQLite database via SQLCipher	Encrypted, on-device data cache. Guarantees that sensitive consumer details and signatures remain secure at rest.
Cloud Backend	Node.js (NestJS) or Go	Extremely fast, type-safe REST APIs capable of handling real-time status updates and document processing queues.
Data Security	AES-256 encryption at rest TLS 1.3 transit encryption	Meets compliance standards for consumer financial/personal documentation. Protects titles and checks from intercept.
Cloud Hosting	Amazon Web Services (AWS)	Scaleable, SOC-2 compliant secure object buckets (S3) and RDS database instances with 99.99% operational uptime.

3. Core On-Site Buyback Workflows

The application provides a structured, fail-safe wizard that enforces DeMars' audit guidelines for vehicle retitling and delivery transactions:

- **Step 1: Odometer & VIN Scan:** The agent uses the smartphone's camera to scan the vehicle's metal VIN plate. The app runs real-time OCR text recognition and automatically cross-references the VIN against the DeMars schedule. Odometer mileage is logged with a mandatory dashboard photo overlay.
- **Step 2: Condition & Key Checklist:** Agents complete a physical checklist verifying the cosmetic panel conditions, tires, and mechanical status. High-resolution photos are captured for each panel, embedding localized GPS and UTC timestamp metadata to prevent retroactive audit tampering.
- **Step 3: Document Verification:** The agent inspects and scans the customer's vehicle title (verifying signatures, release of bank liens, and power of attorney documents) and takes photographic proof of the check handed to the customer.
- **Step 4: Dual E-Signatures & Checksum Seal:** Once verification is completed, both the Transfer Agent and the customer sign the digital pads. The app captures coordinate points, signs the forms, and seals the record with a cryptographically secure SHA-256 hash.

4. Security & Cryptographic Integrity Controls

- **Offline-First SQL Sync:** Transfer Agents regularly work in dealership basements, steel-frame lots, or remote areas. The platform securely caches all forms, photographs, and signatures in a local SQL database. Once a web connection is recovered, background sync tasks push logs back to DeMars cloud storage.
- **SHA-256 Handover Checksums:** To satisfy manufacturer legal requirements, the application generates a local SHA-256 cryptographic hash of all combined data (VIN, odometer, photos, signatures) on submission. This checksum acts as a digital tamper-evident seal.
- **GPS Geofencing Validation:** The app records the device's precise GPS coordinate bounds at the time of e-signature, legally verifying that the transaction occurred at the correct dealership location.
- **Encrypted HQ Messenger:** A secure messaging gateway connecting agents directly to DeMars support. If a lien dispute or check discrepancy arises, agents can text support, upload documents, and get real-time clearance.

5. Project Implementation Roadmap

GS3 Solutions proposes an agile 12-week development lifecycle to design, build, and deploy the application:

Development Phase	Timeline	Core Activities
Phase 1: UX/UI & Workflows	Weeks 1 - 2	Detailed wireframing of Transfer Agent screens. Alignment on Lemon Law and buyback legal requirements. Interactive mockups creation.
Phase 2: Mobile App Shell	Weeks 3 - 5	Setting up React Native / Flutter framework. Structuring local SQLite schemas with SQLCipher database encryption.
Phase 3: APIs & Sync Engines	Weeks 6 - 8	Building Node.js cloud APIs, AWS S3 image upload tunnels, and robust offline sync queue managers.
Phase 4: Integrations & OCR	Weeks 9 - 10	Integrating OCR camera parsing for VINs, building signature pads, and completing security penetration audits.
Phase 5: UAT & App Store Launch	Weeks 11 - 12	User acceptance testing with field agents. Finalizing compliance checks and submitting to Apple App Store and Google Play.

6. Post-Launch Maintenance & Support SLA

To ensure complete reliability in operational environments, GS3 Solutions provides continuous support:

- **Platform OS Upgrades:** Annual updates for iOS and Android, ensuring the application remains backward compatible and functional with new mobile releases.
- **Database Security Audits:** Biannual checks of SQLite and AWS server policies to prevent data leakage and maintain security standards.
- **24/7 Field Support Hotline:** A dedicated support line for Transfer Agents who encounter critical, blocker errors while executing vehicle buybacks on-site.

Gisidhar Datta
Director